

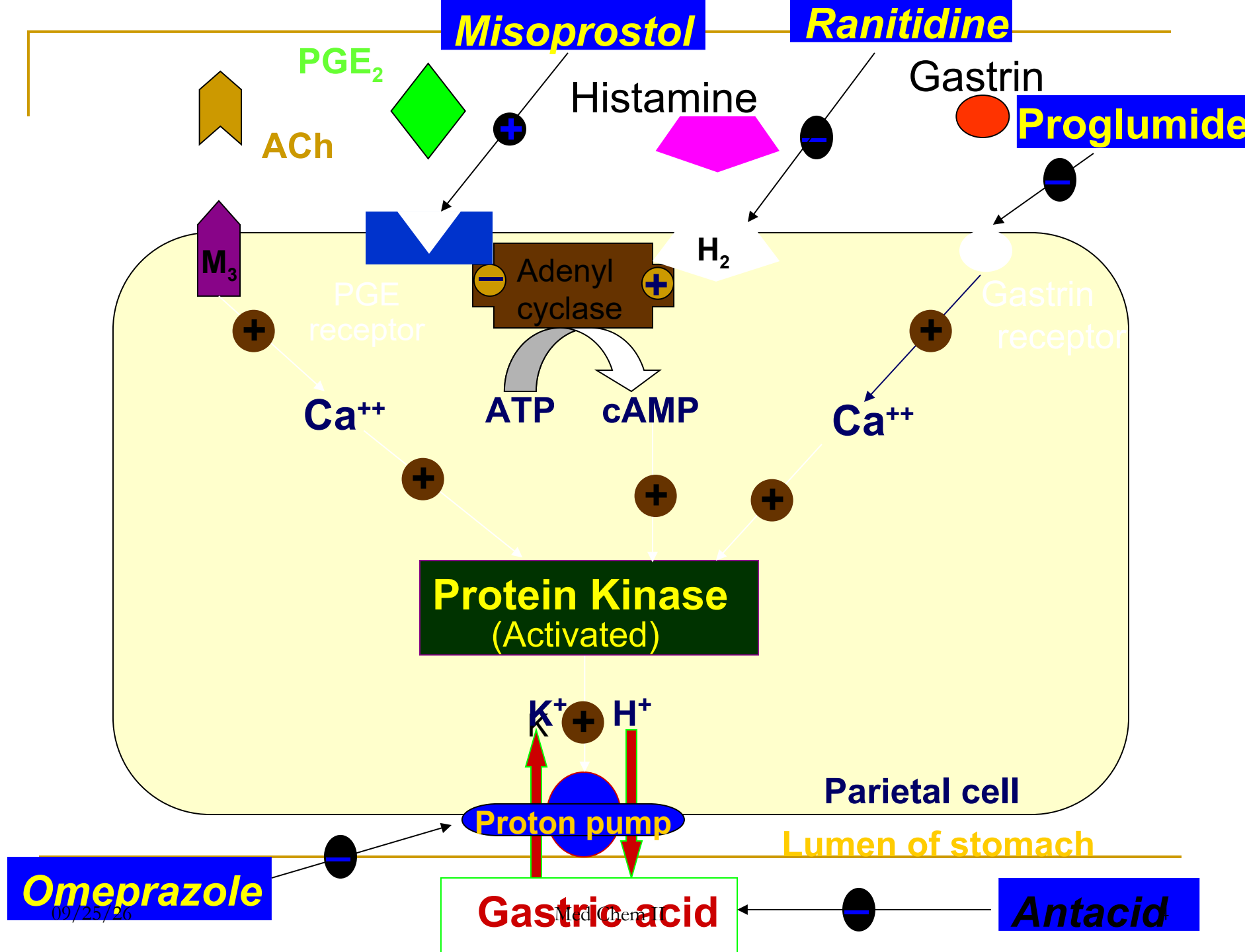
Gastrointestinal Agents

By Yitagesu T.(M.Sc)

-
- Drugs used for treatment of GI disorders includes
 - A. Antiulcer agents
 - B. Drugs used for treatment of constipation and diarrhea
 - C. Emetics and Antiemetics
 - D. Drugs used for inflammatory bowel disease

A. Drugs used for treatment of peptic ulcer

- Peptic ulcers are localized erosions of the mucous membranes of the stomach or duodenum
- Causes:
 - Increased acid secretion
 - H.pylori
 - Decrease in mucosal defense (usually by NSAIDs)



- Drugs used for treatment of peptic ulcer includes
 - Antacids
 - H₂-receptor antagonists
 - Proton pump inhibitors
 - Cytoprotectants and adsorbents

1. Antacids

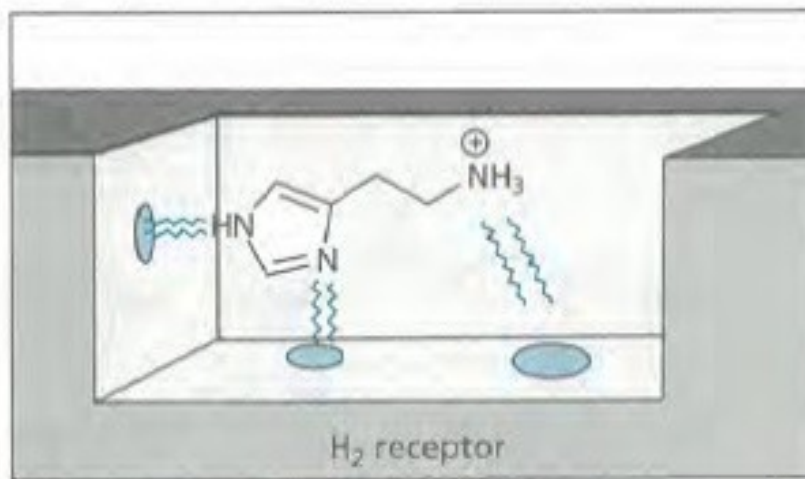
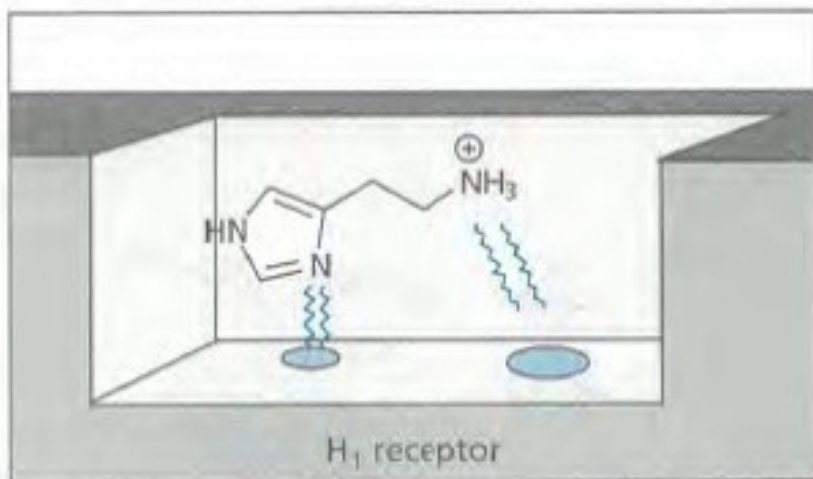
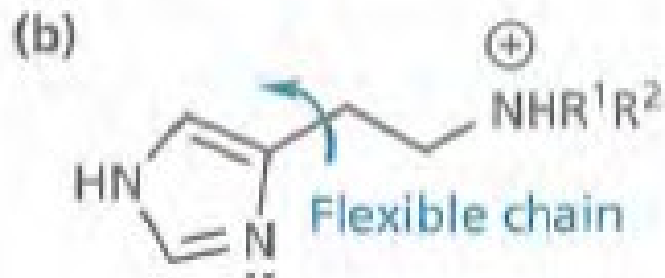
- The most basic therapy for gastric hyperacidity is the use of basic inorganic salts as **antacids**. There are a variety of these agents available for over the counter use.
- Antacids are classified by their cationic metal, usually a sodium, magnesium or aluminum.
- Sodium antacids should be avoided in hypertensive patients, magnesium antacids can cause diarrhea, and that aluminum antacids can cause constipation.

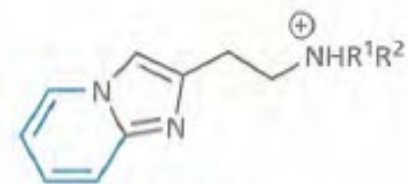
2. H₂ Receptor Antagonists

- Early attempts to develop a histamine antagonist specific for the H₂ receptor were focused on the observation that histamine was potent stimulus for gastric acid production *in vitro*

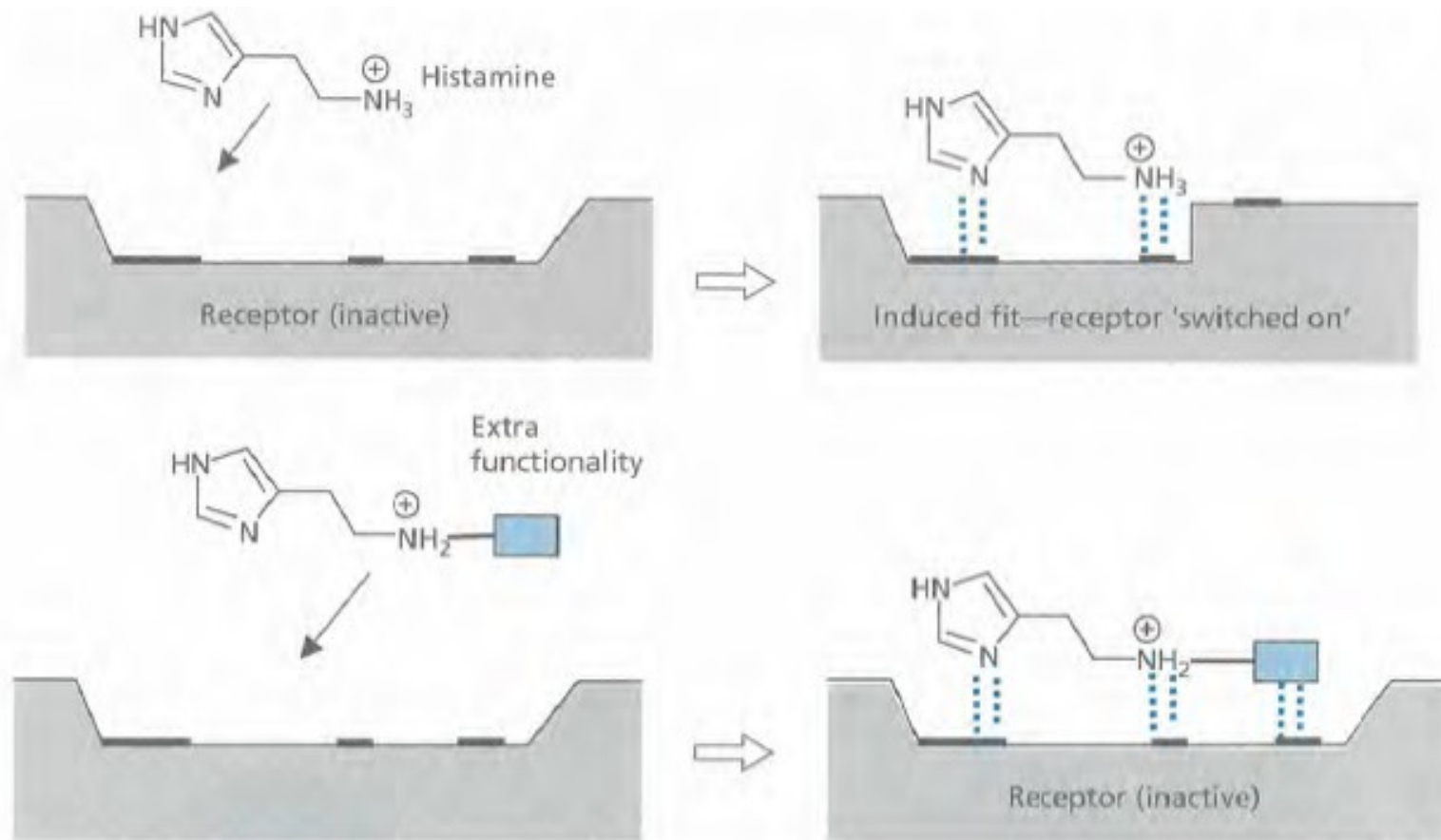


SAR of H₁ and H₂ agonists

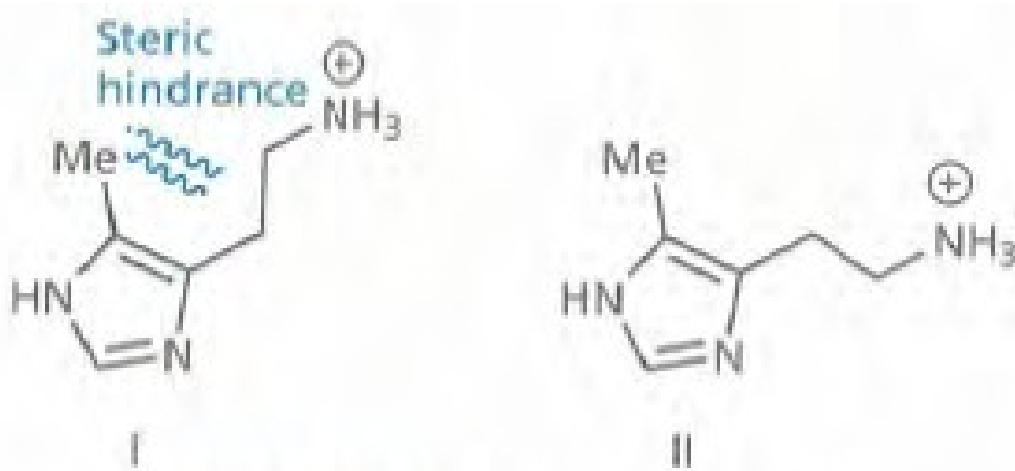




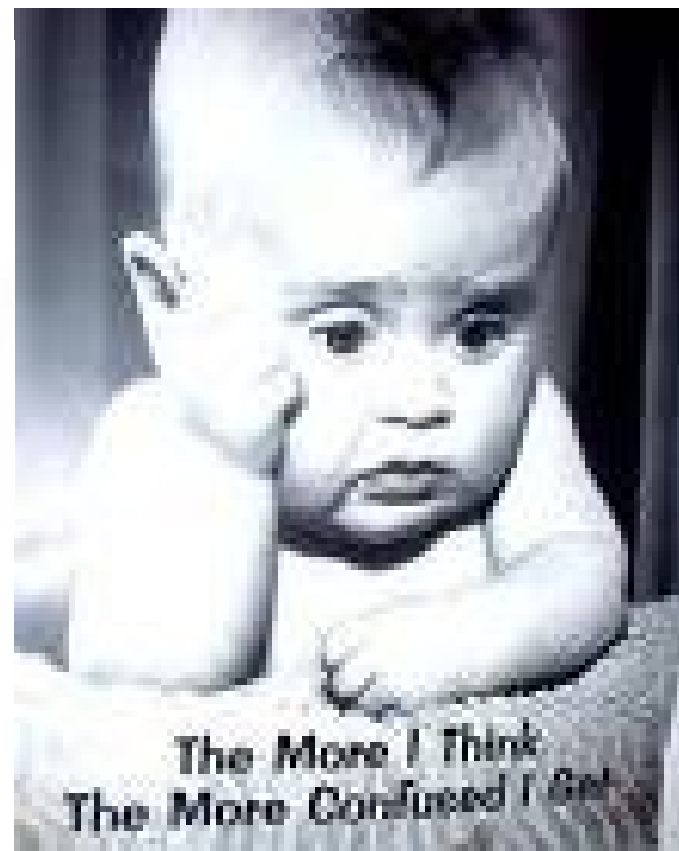
Early unsuccessful attempts to develop H₂ antagonists...



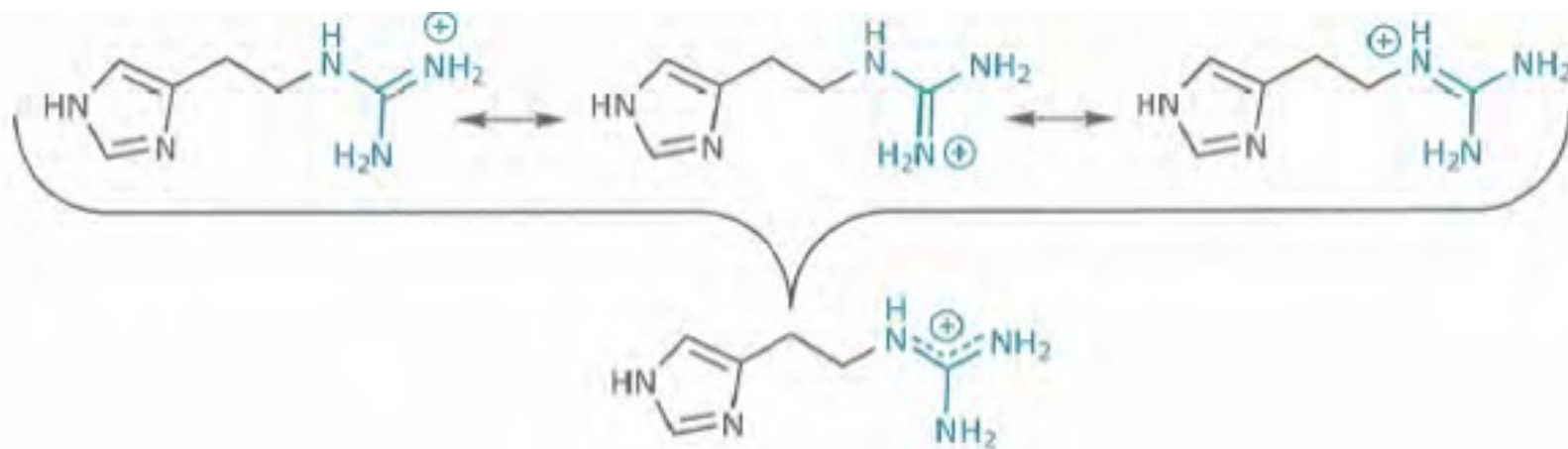
Discovery of H₂ agonist...4-methylhistamine...



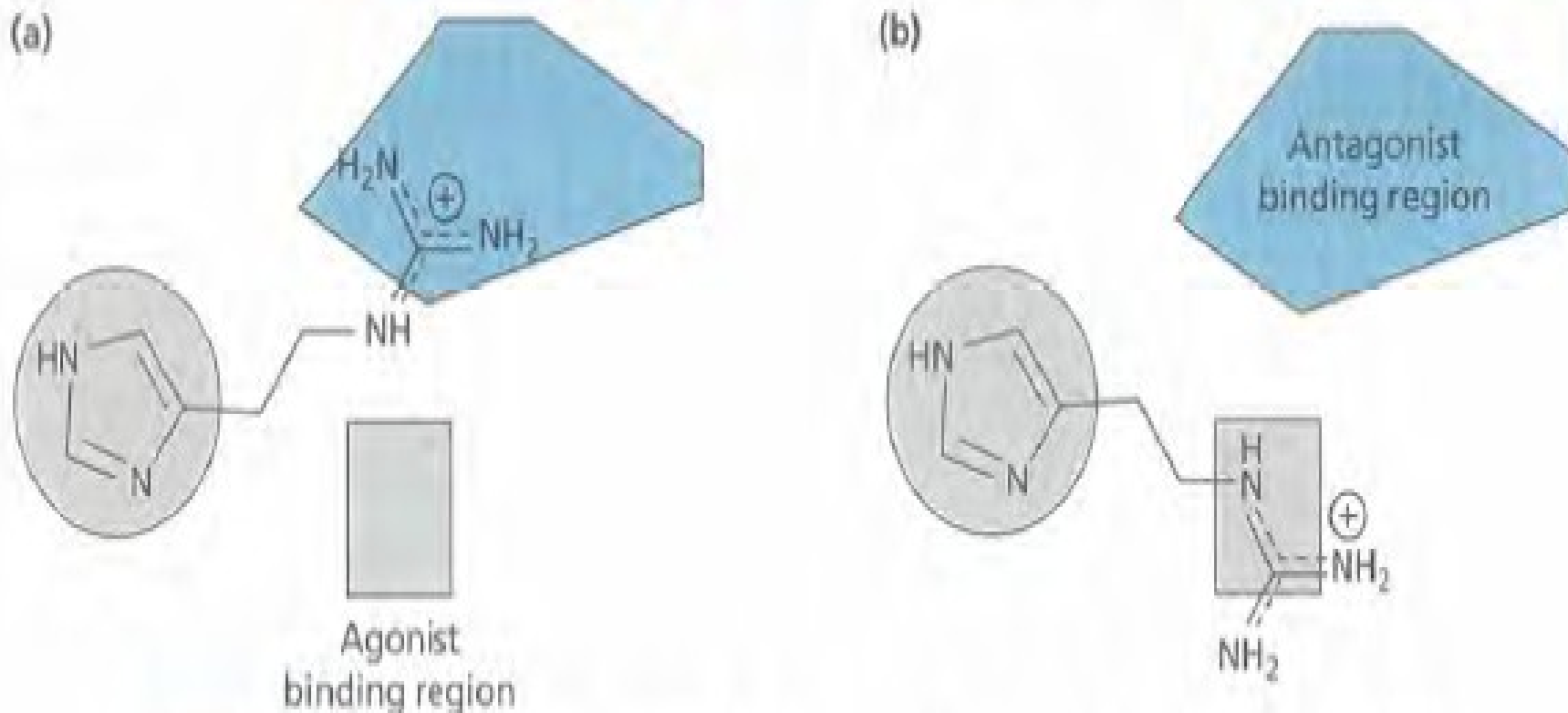
- Further 200 cpds were synthesized but non of them were active



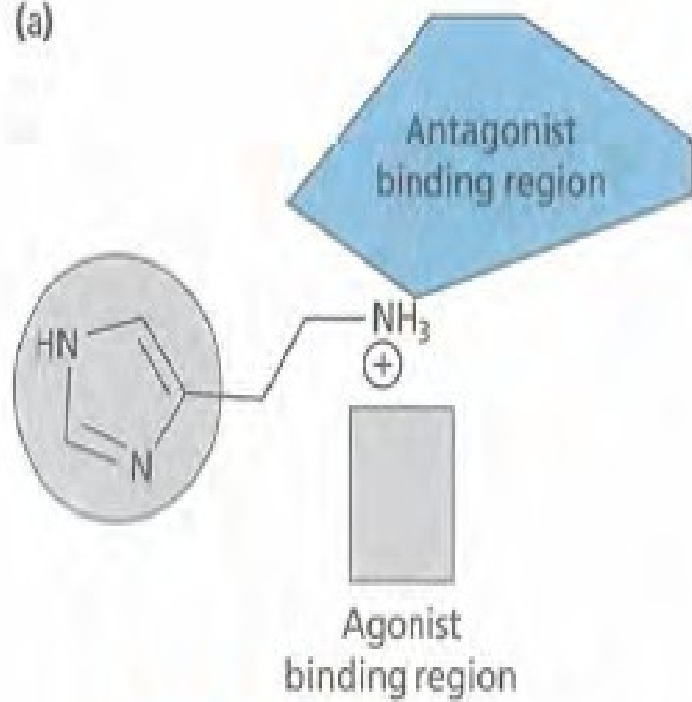
- Focus switched to study of the effect of varying polar groups on the molecule...
- And this resulted in the discovery of N-Guanylhistamine...a weak antagonist of gastric acid release [crucial breakthrough]



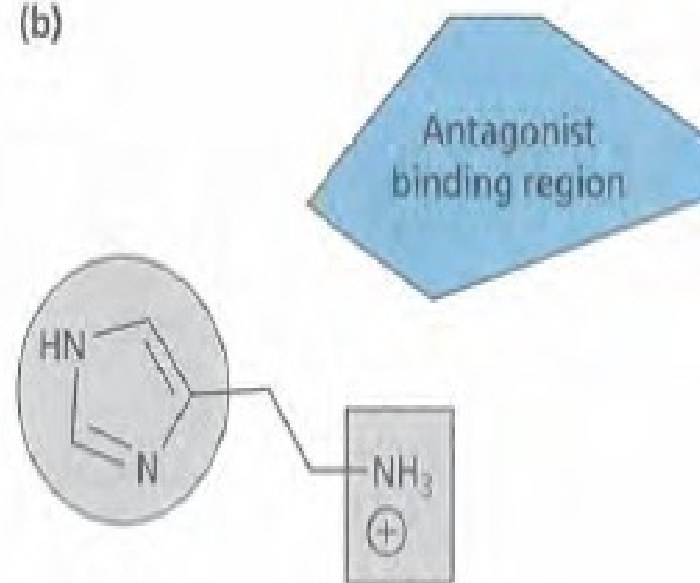
Proposed binding....



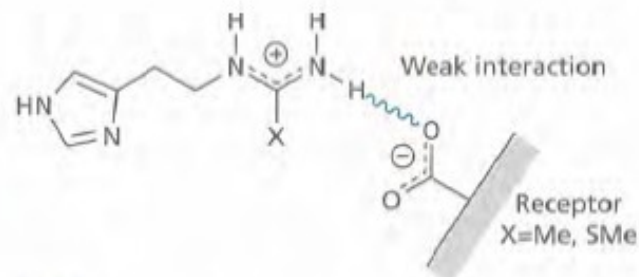
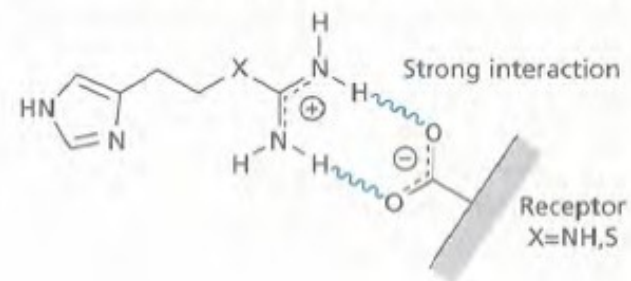
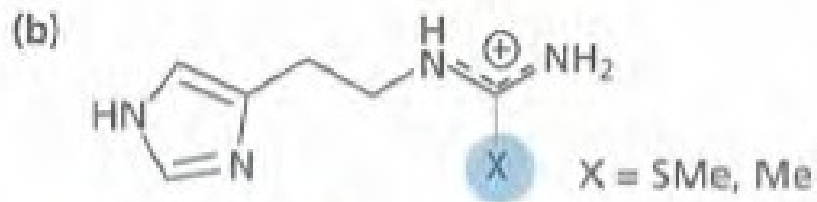
(a)



(b)



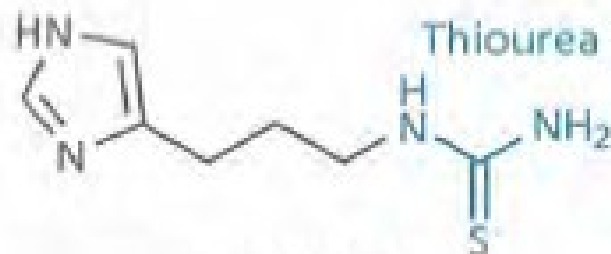
Developing the lead...a chelation bonding theory



- Conclusion: the two terminal nitrogen are important

From partial agonist to antagonist...

- Development of burimamide...by avoiding the +ve charge



SKF 91581

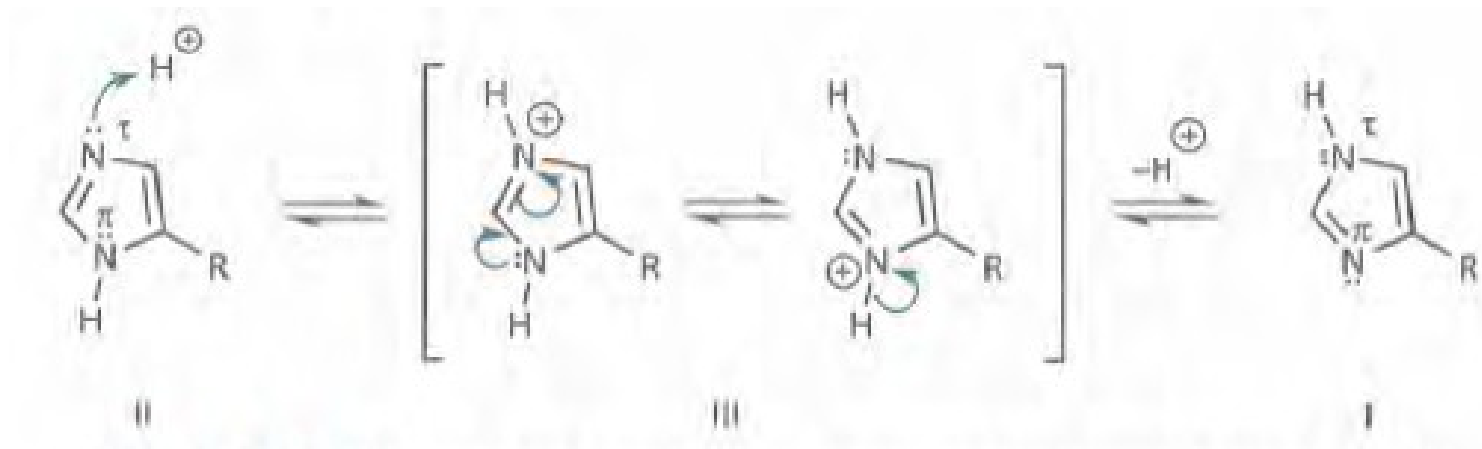


Burimamide

- Burimamide was highly competitive antagonist at H-2 receptor.
- Was 100x more potent than N-guanylhistamine derivatives in inhibiting histamine mediated gastric acid release.
- Its discovery gave SKF research team groups far more strong evidence for the existence of H2 receptors
- But despite these success burimamide was not suitable for clinical trials since its activity was still too low for oral administration.

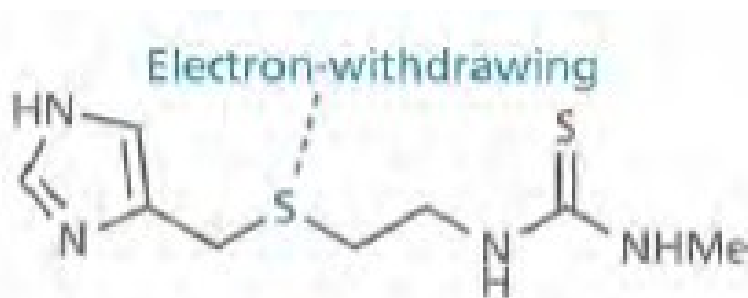
Development of metiamide...

- Attention was now diverted to the imidazole ring again to increase the activity of burimamide

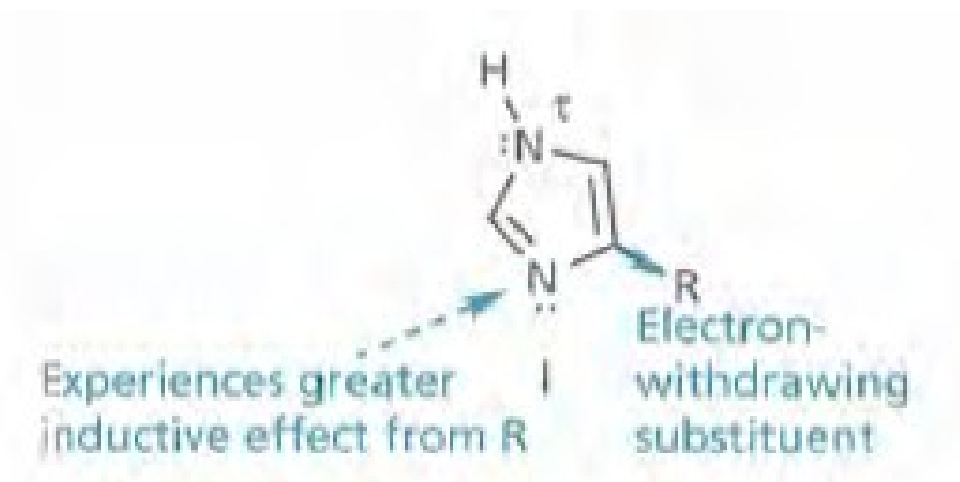


Imidazole ring can equilibrate b/n tautomeric forms (I and II) via the protonated intermediate (III)

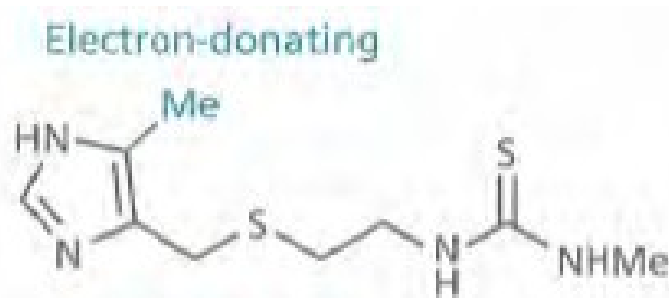
- pka for imidazole ring alone = 6.80
- pka for imidazole in histamine = 5.74
- pka for imidazole in burimamide = 7.25
- It might be protonated at physiological PH and this might affect the binding mode of burimamide with the receptor
- To tackle this problem, electron withdrawing group was introduced.... **Thiaborimamide**



- The next step was to make the imidazole ring similar to histamine by increasing the basic nature of N- τ by increasing the basic nature of N- τ



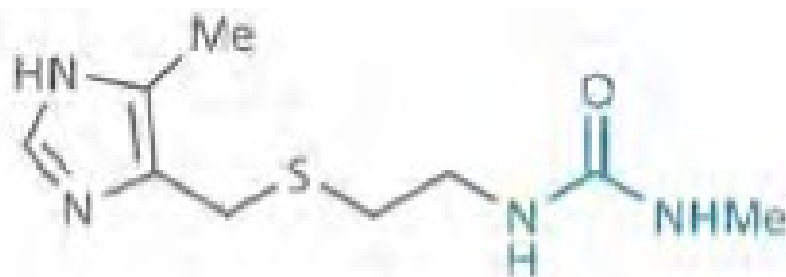
- This can be done by inserting methyl group...which gave rise **metiamide**



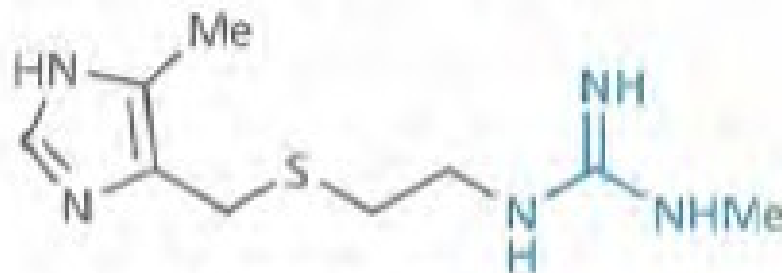
Metiamide

- Was 10X more active than burimamide in its antagonistic effect
- But was shown to be very toxic (nephrotoxicity and granulocytopenia). Hence, was not marketed
- The toxicity was attributed to the **thiourea group**
- So the next attempt was to change this group

- Two unsuccessful attempts

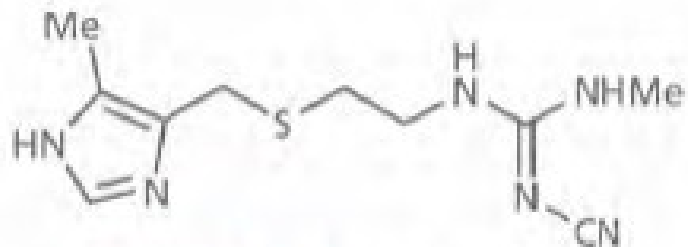


Urea analogue

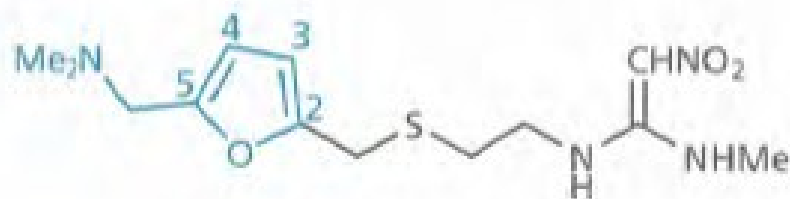


Guanidine analogue

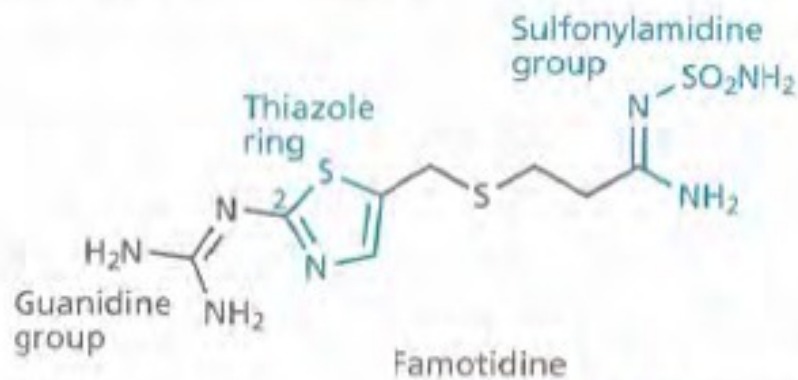
Finally cimetidine and other derivatives....



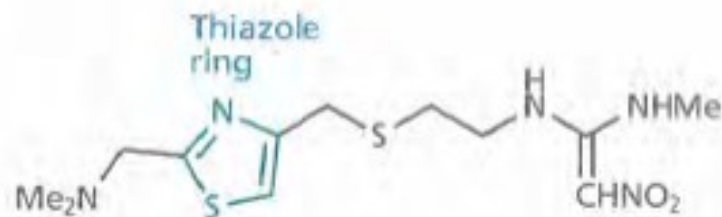
Cimetidine



Ranitidine

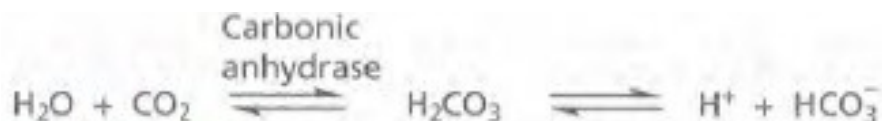
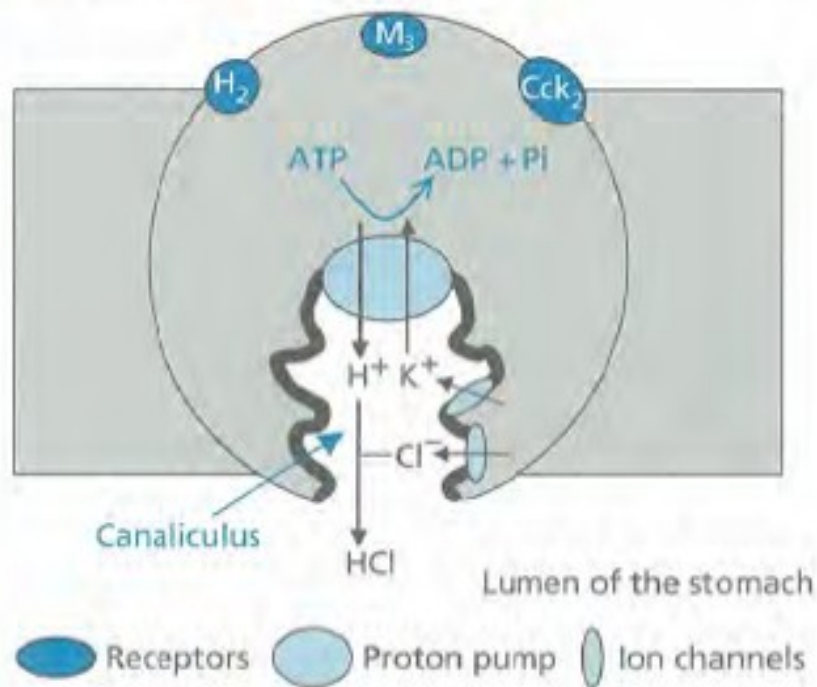


Famotidine



Nizatidine

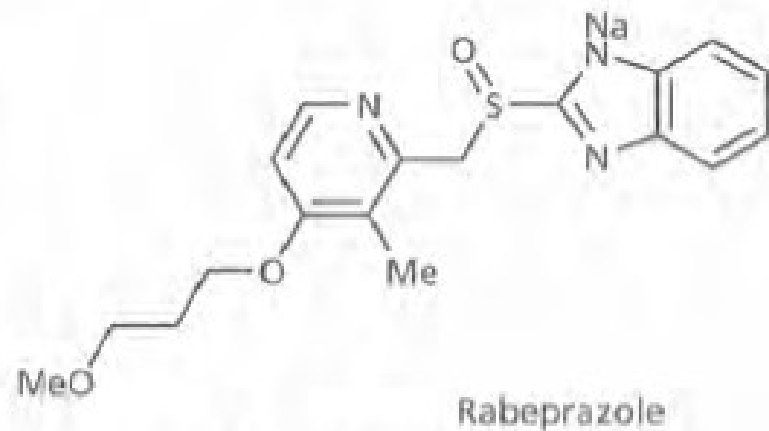
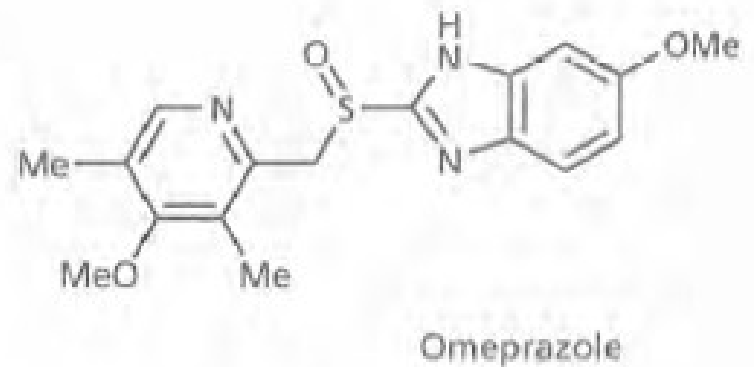
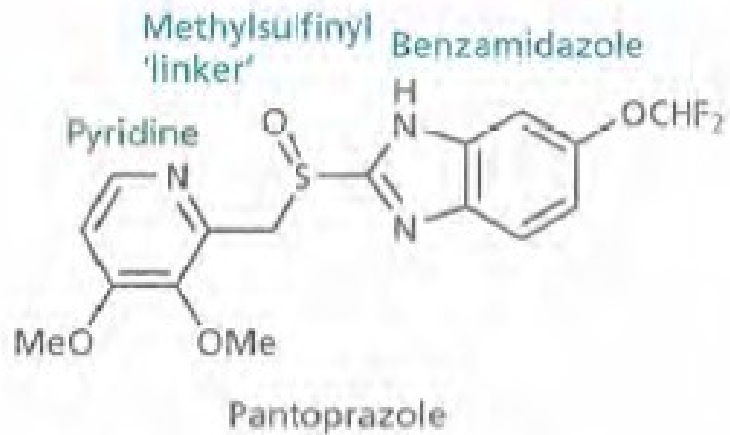
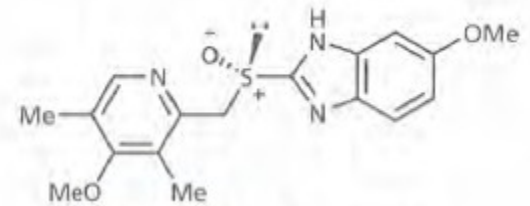
3. Proton Pump Inhibitors



- The source of proton is bicarbonate which is produced in the cells from CO₂ and H₂O by carbonic anhydrase
- The pumping process is energy dependant since both ions moves against their conc.grediant.

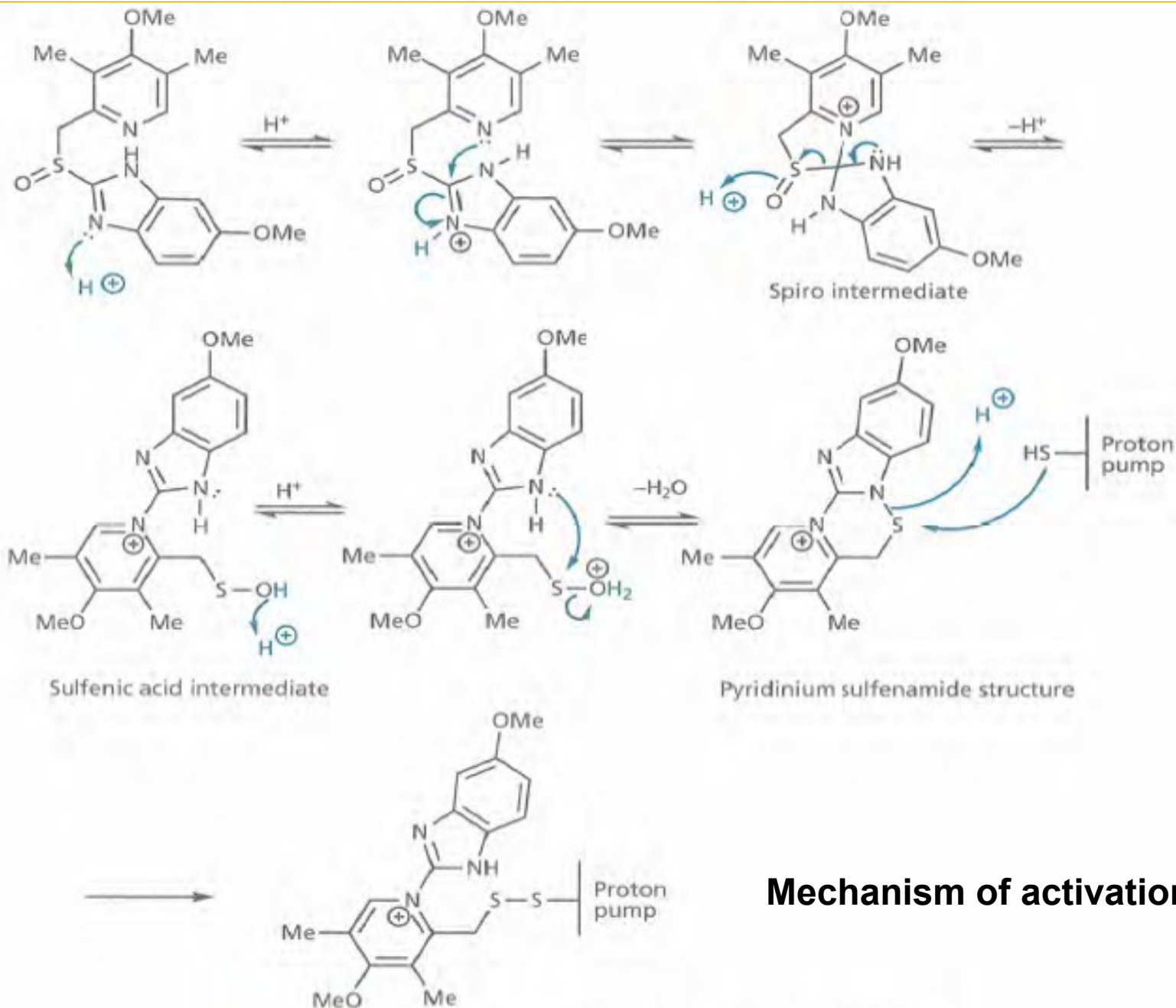
Proton pump inhibitors (PPIs)

- There are 5 PPIs currently on market which includes omeprazole, esmaprazole, lansoprazole, **pantoprazole** and rabeprazole
- All the PPIs have pyridyl methylsulfinyl benzamidazole skeleton and act as **pro-drugs**, as they are activated when they reach the **acidic canaliculi of parietal cells**.



Mechanism of inhibition

- The PPIs are weak bases having pK_a of 4.0
- As a result they are weak bases at the blood pH and are only ionized in strongly acidic environment where the pH is less than 4.
- These are conditions found only in the secretory canaliculus of the parietal cells, where the pH is 2 or less
- Activation of the drug occurs at the **canaliculus** and the drug will **irreversibly forms covalent bond with the cystine amino acid residues found on the proton pump.**



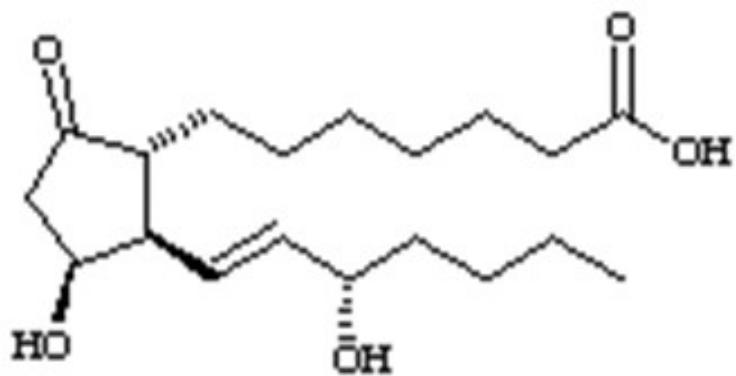
Mechanism of activation....

4. Cytoprotective Agents and adsorbents

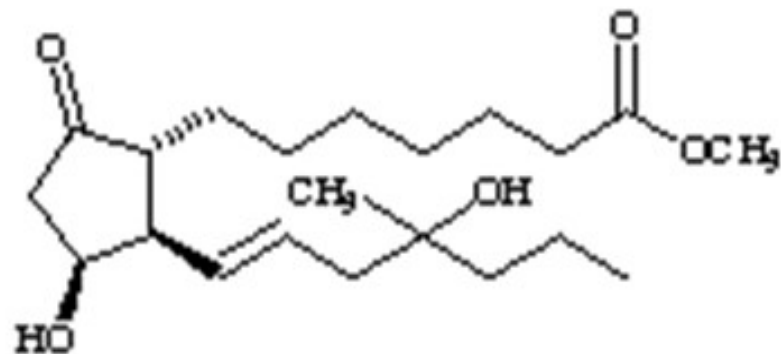
- It has been known for a number of years that prostaglandin synthetase is present in the gut wall,
 - And those prostaglandins of the E and F series are synthesized and secreted in gastric juice.
 - **Prostaglandins (and in particular PG_E) can inhibit histamine-stimulated HCl release without interrupting mucosal flow.**
 - And are thus capable of protecting the mucosa, a phenomenon termed **cytoprotection**.

- Other effects include enhancement of mucosal blood flow, stimulation of bicarbonate release and increased mucus production.
- Because prostaglandins are short-lived intermediates *in vivo*, it is generally not possible to use these agents as cytoprotectants.
- However, there are now a number of analogues of PGE₁ which are useful as cytoprotectants.
- The stable analogue **misoprostol** was recently marketed for this purpose.
- Misoprostol has two structural features which make it more stable to metabolism than PGE₁:

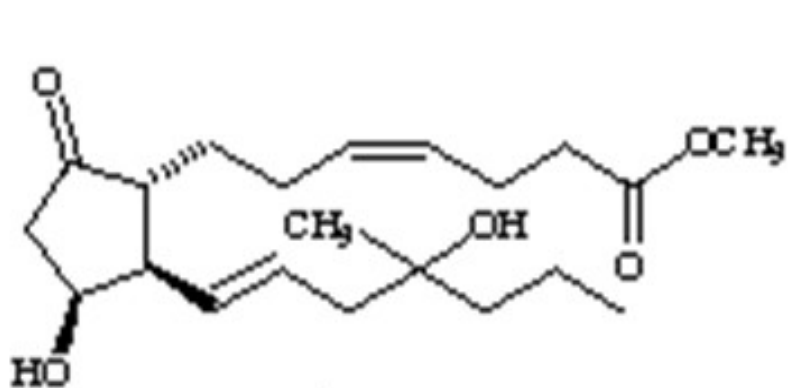
- The hydroxyl substituent is moved from the 15 to the 16 position, where it resides geminal to a methyl group, greatly reducing metabolism by oxidation, and it is administered as the methyl ester.
- The methyl ester is a pro-drug form which must be cleaved prior to activity. Misoprostol retains all of the cytoprotective attributes of PGE1.
- Additional cytoprotective agents related to misoprostol are in development, as shown below.



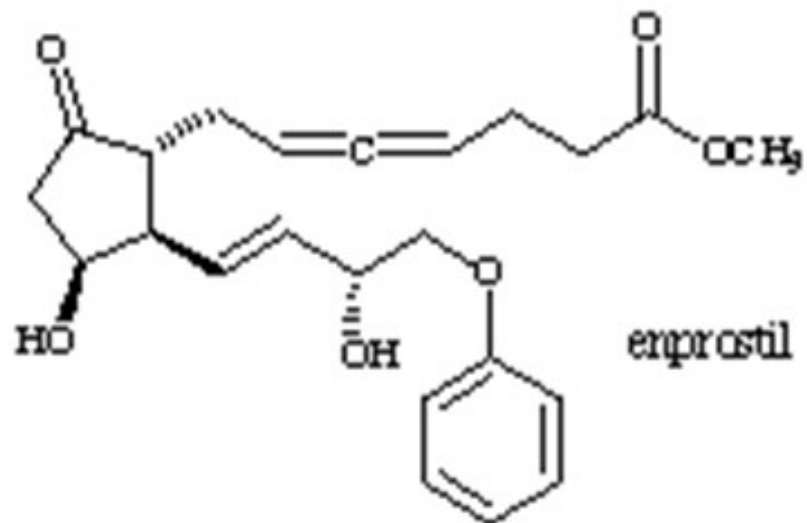
prostaglandin E1



misoprostol

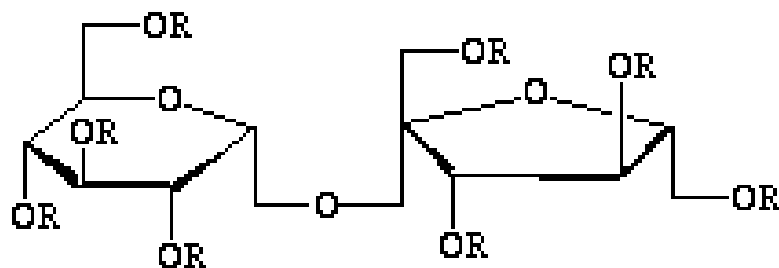


enisoprost



enprostil

- There is one additional cytoprotective agent which acts by a completely different mechanism.
- **Sucralfate (Carafate®)** protects the gastric mucosa by forming a sticky, viscous gel that adheres to mucosal tissue and protects it from the action of gastric acid and pepsin.
- Sucralfate has no inherent acid-neutralizing activity.

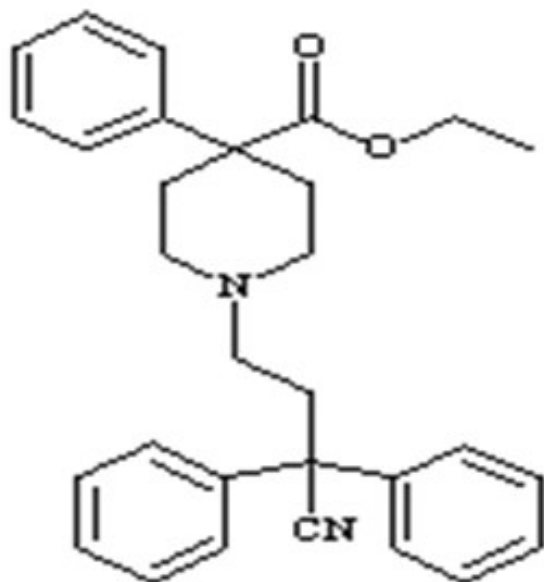


sucralfate
(Carafate®)

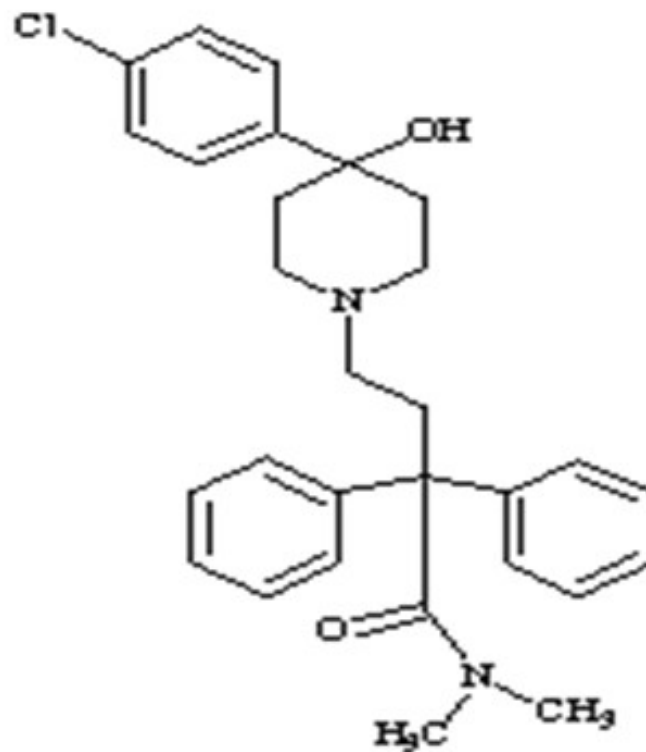


B. Antidiarrheals

- Commonly used antidiarrheals include the meperidine derivatives **diphenoxylate and loperamide**.
- Both are structurally related to 4-phenylpiperidine opioid analgesics, but have negligible opiate effects and a low abuse potential.
- These agents act by reducing the propulsive activity of the intestinal tract, a common side effect of the opiate analgetic agents.
- In fact, codeine may also be used for this purpose, but is less frequently used due to its **addiction liability** and **potential for abuse**.



difenoxylate

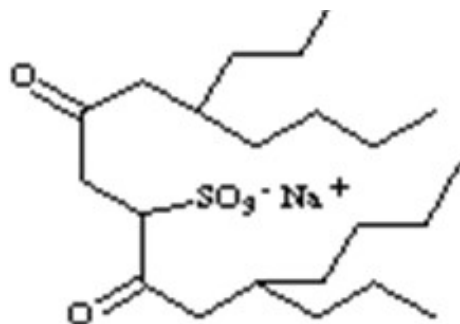


loperamide

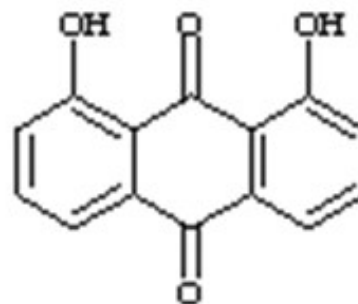
C. Laxatives

- Laxatives fall into one of several types:
- **Surfactants** act by a detergent effect which enhances the mixing of fat and water to soften stools.
 - The prime example of a surfactant laxative is **dioctyl sodium sulfosuccinate**.
- **Irritant** laxatives act by direct contact with the gastric mucosa, and the resulting irritation causes a decrease in transit time.
 - Examples of irritants are **danthron, bisacodyl, phenolphthalein, castor oil, and senna.**

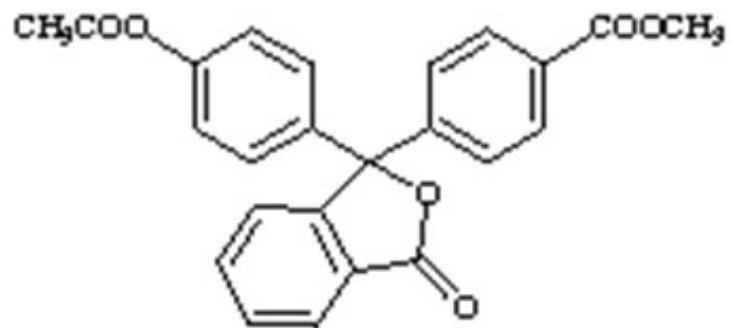
- **Bulk** laxative act by increasing stool volume, and include psyllium-based laxatives such as **Metamucil®**, methylcellulose containing laxatives and polycarbophil.
 - **Lactulose**, a simple disaccharide, is metabolized by colonic bacteria to lactate and small amounts of acetic and formic acids,
 - Resulting in acidification of the colonic contents and a resultant catharsis.



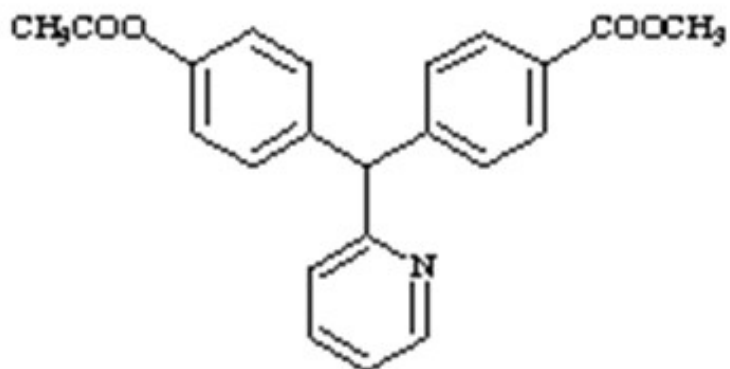
dioctyl sodium sulfosuccinate



anthracene



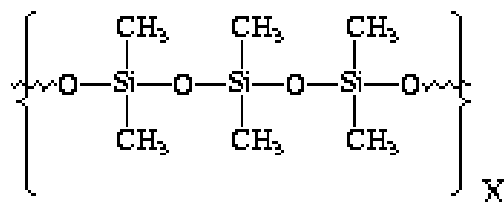
phenolphthalein



bisacodyl

Antiflatulent Agents

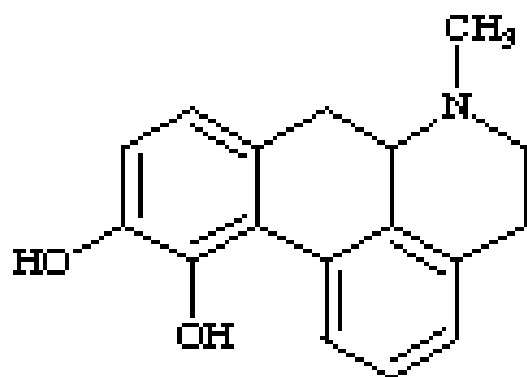
- **Simethicone** is the most commonly used antiflatulent, and
- Acts by dispersing excess gas in the intestine before the patient has the opportunity to do so.
- It is a simple, methylated siloxane. Peppermint spirit and activated charcoal are also useful in this regard, but are less well tolerated.



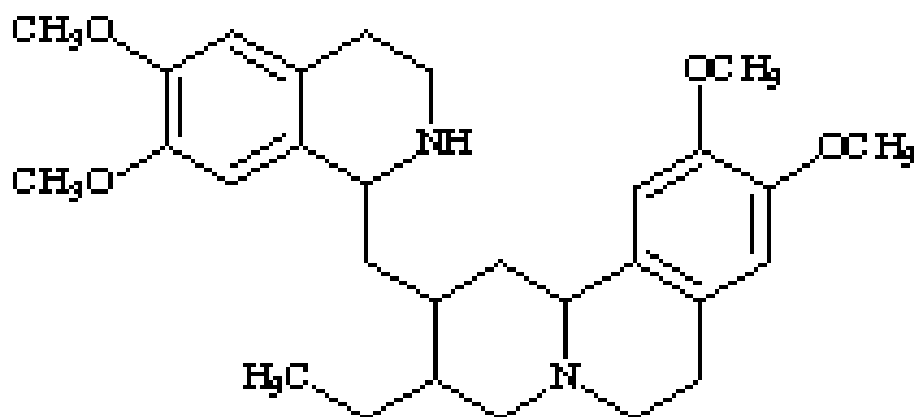
simethicone

D. Emetics and Antiemetics

- A limited number of drugs are available for induction of vomiting in patients who have ingested toxic materials.
- The morphine analogue **apomorphine** can be used intravenously for this purpose.
- In addition, **syrup of ipecac**, a solution of plant alkaloids containing the emetic compound **emetine**, is used orally to induce emesis.
- Both of these drugs act by stimulation of the stomach and a direct effect on the chemotrigger zone.



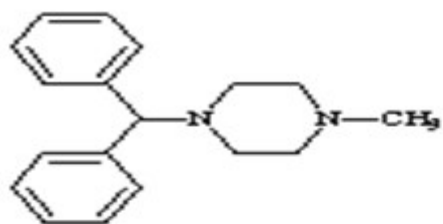
apomorphine



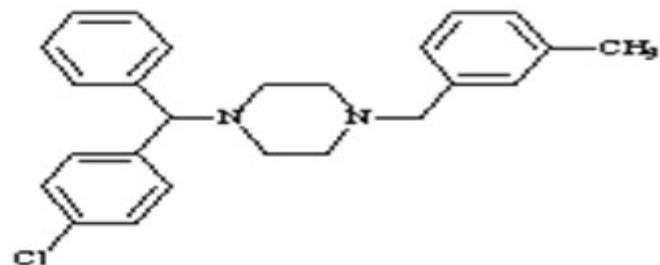
emetine

- A variety of phenothiazine and related analogues are useful for the treatment of nausea, and have a pronounced antiemetic effect.
- The phenothiazines have a direct effect on the chemotrigger zone in the CNS.
- **Prochlorperazine** is primarily used for this effect, since it has minimal neuroleptic activity.

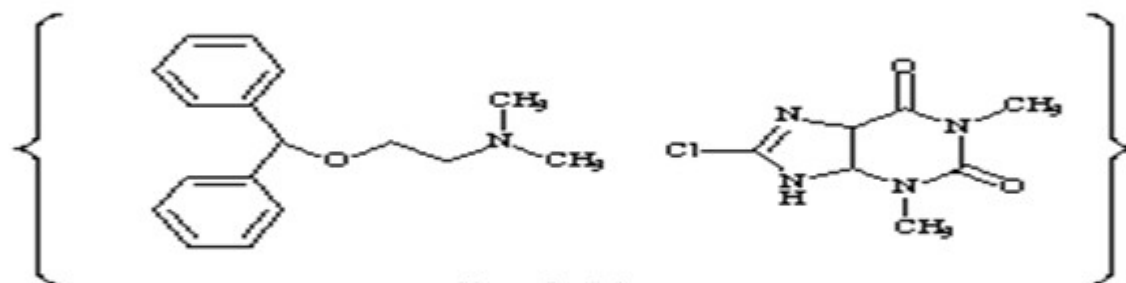
- Prochlorperazine is not effective for preventing nausea due to motion sickness.
- Other compounds are more often used for this syndrome, including **meclizine, dimenhydrinate and cyclizine**, again due to their effect on the chemotrigger zone.
- Scopolamine patches are also effective in the relief of motion sickness.



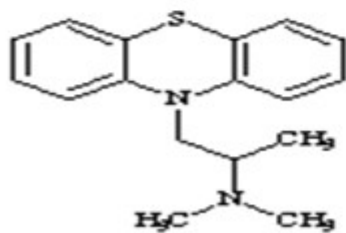
cyclizine



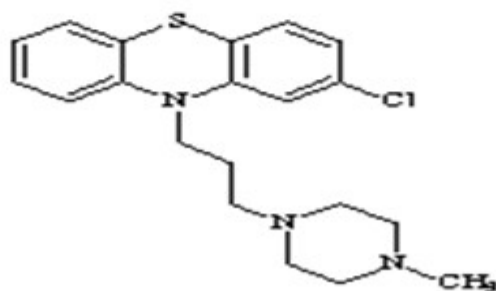
meclizine



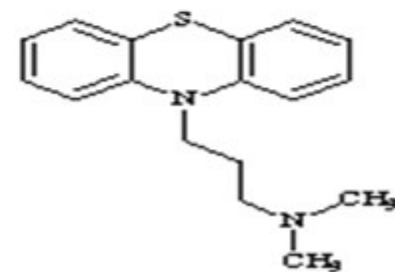
dimenhydrinate



promethazine



prochlorperazine

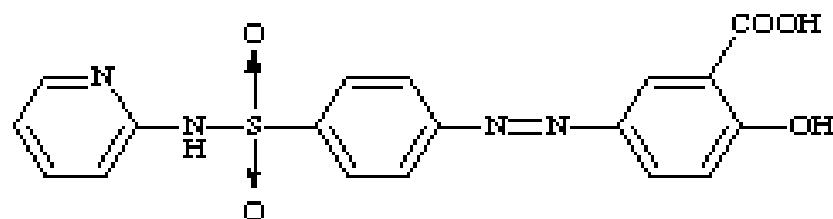


chlorpromazine

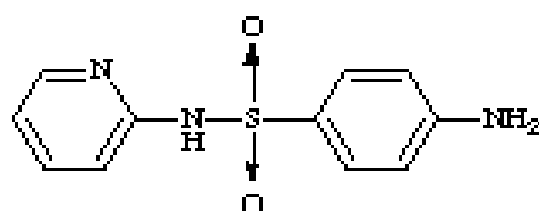
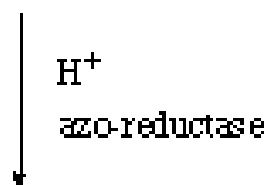
E. Drugs Used in the Treatment of Inflammatory Bowel Disease

- The sulfonamide derivative **sulfasalazine** is widely used for prolonged treatment of ulcerative colitis.
- The drug is practically insoluble in water, and as such is poorly absorbed from the intestine and reaches the colon intact.
- In this location, bacterial azo-reductase enzymes cleave sulfasalazine to sulfapyridine and **5-aminosalicylic acid**, which appears to be the active constituent.

- The mechanism of sulfasalazine involves the 5-aminosalicylic acid-mediated inhibition of prostaglandin synthesis in the colon, thus relieving the resultant inflammation.
- 5-Aminosalicylic acid is effective when administered alone, and is marketed separately as mesalamine.

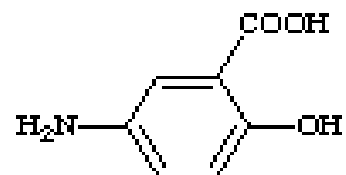


sulfasalazine
(Azulfidine®)



sulfapyridine

+



5-aminosalicylic acid
mesalamine
(Pentasa®, Asacol®,
Roxasa®)